
CSCI 4193/5193 | Winter 2026

Technology Innovation

Tuesdays 1435 -1725

Rowe Management Building Room 1024

Technology innovation combines design thinking, rapid prototyping, and software development to validate ideas for new technologies and business models. Students work in interdisciplinary teams to address a design challenge. The ideas that teams advance will reflect the powerful synergies between software development, design thinking and entrepreneurship.

RATIONALE

The job market increasingly emphasizes adaptability, customer empathy, critical evaluation, and collaboration. We practice these high-performing skills while directly experiencing the innovation lifecycle of value creation, value capture, and value exchange.

COURSE LEADERSHIP

Instructor Andrew Cochran - ac@dal.ca (**include CSCI x193 in the subject line**)

Office hours Book time using the calendar tool on MS Teams
Meet at the Goldberg CS Building, Room 308

IMPORTANT DATES

- First class meeting - 13 January
- Last day to change and add classes - 22 January
- Last day to drop winter courses without a "W" - 05 February
- Munro Day (university closed) - 06 February
- Nova Scotia Heritage Day (university closed) - 16 February
- Winter Study Break (no classes) - 16-20 February
- Last day to drop with a "W" - 09 March
- Good Friday (university closed) - 03 April
- Exam period - 11 April - 27 April

ASSESSMENT

Course assessment methods combine individual and team work, assignments, quizzes, and examinations, broken down as follows:

■ Assignments (23%)

See Table 1 for dates, tasks, and values.

■ Weekly quizzes (10%)

The best eight results, worth 1.25% each

■ Examinations (65%)

There are two examinations, one at the midterm (optional) and the final at the end of the term. The questions will cover all content to date in the class meetings, the associated readings, and the assignments. Both exams will be closed-book, written in provided booklets, and in-person only. You will need to show a Dal photo ID to submit each exam.

- **The midterm exam (15%)** will be held during the class meeting on **February 24**. The midterm is optional for students wishing to measure their understanding of the material to date and practice for the final examination. Students choosing not to take the midterm will have its value attached to their final examination, provided they give written notice to the instructor on/before February 17 using the designated dropbox on Brightspace.
- **The final exam (50%)** will be during the final exam period, April 9-26. The registrar's office will post the day, time, and location. Students must be present in person on the day and time of the final examination.

Table 1: Dates, tasks, and values for the individual and team assignments.

Assignment	Done by	Due date	Task	Value
Project journals - interim (written)	Individual	Feb 10	Showing familiarity with Project Journals	1%
Product ideas & support (written)*	Team	Feb 13	Proposing products that address users' needs	1%
Prototype plan & support (written)*	Team	Mar 06	Having a clear vision for a product to create	3%
Product. pitch, & materials (written & spoken)*	Team	Mar 30 (spoken live on Mar 31)	Presenting an investor-ready pitch as a team	3%
Project journals - final (written)	Individual	Mar 31	Providing evidence of originality	4%
Individual contribution to product, pitch, & materials (written & spoken)	Individual	April 2	Contributing to the product & pitch as an individual	7%

Shiftkey reflection (written)	Individual	Apr 09	Identifying innovation values in 3rd party presentations	4%
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** These submissions will combine into a single letter grade for your work as a team for the term*

*** Team presentations will be on March 31 with the order to be determined by in-class lottery*

■ Course activities (2%)

Students will receive points from contributing to various activities throughout the term, to a maximum value of 2%, from among the following:

- Posting original day one project - .5
- Submitting relevant & reliable news articles about innovation - .5
- Doing a team standup - .5
- Submitting original product feedback - .5
- Submitting original pitch feedback - .5

COURSE POLICIES

A. Requirements to pass

1. **For 4193 students:** A minimum of D is required to pass this course. To obtain a D grade, the aggregate of all course assessments must be at least 50% of their stated value, and you also must achieve at least 50% in the aggregate of the midterm and final exam; if choosing not to take the midterm, this must be achieved in the final exam.
2. **For 5193 students:** A minimum of B- grade is required to pass this course. To obtain a B- grade, the aggregate of all course assessments must be at least 70% of their stated value, and you also must achieve at least 70% in the aggregate of the midterm and final exam; if choosing not to take the midterm, this must be achieved in the final exam.
3. The grade conversion scale in Section 17.1 of the Academic Regulations will be used. See [Grade Scale and Definitions](#).
4. All end-of-term grades are final.

B. Attendance

1. This course is conducted in person. You are expected to attend all classes in person unless you are ill.
2. No accommodation is provided to students who are not in Halifax.

C. Submissions

1. All submissions for grading are due on the day and time indicated.
2. There are no extensions for work to be graded.
3. If a submission is only partially complete at the deadline, submit the incomplete work.

4. A grace period of 31 minutes is provided to allow for technical mishaps when uploading submissions for grading. Those submitted after the expiry of the grace period (I.e., 2359 + 31 minutes = 12:30 a.m. AT) will not be graded, and the mark will show as 0/F.
5. Only your most recent submission before the deadline will be graded.
6. Submissions for grading will be filtered through similarity detection software.
7. Any submission for grading may be subject to additional review. The results may not be published simultaneously with others.
8. A submission for grading that is sent for an academic integrity review will be published after the academic integrity finding.

D. Accommodations

1. Accommodations must be arranged in advance of the time or event that will be affected. They are impossible afterwards.
2. A Student Declaration of Absence (SDA) may be used when you cannot attend a class. Use the [SDA Form](#). An SDA must be submitted before the class that will be missed. Submit the completed SDA form via *Brightspace > This Course > Assessments > Assignments > Student Declaration of Absence*. Note:
 - SDAs replace sick notes and cannot be used in addition to absence for illness.
 - SDAs cannot be used to create an extension for an assignment.
 - Up to two SDAs may be used in a term.

E. Generative AI and other assistive software

1. You may use generative AI and other assistive software tools subject to the provisions of each assessment. While we expect you to use these tools for learning, specific assessments in this course will disallow the use of AI-driven tools to assure that you have attained course learning outcomes. This is because a graduate must be able to analyze, assess and produce work unassisted by AI technology. Examples of generative AI include diffusion models, for image generation and video/image synthesis, and large language models, for text generation and content/code writing. Examples of assistive software are citation generators, spelling and grammar checkers, and templates.
2. Particular standards for each submission, including the use, treatment, or attribution of generative AI material, will be included in each assessment's written guidelines and are final. If the work you submit is believed to misrepresent your understanding or competencies, you may be asked to complete a supplemental assessment to evaluate your mastery of the course objectives.

F. Regrades

1. You may ask for a re-grade for any assignment. Regrades are a new and complete assessment of your original submission, done by the course instructor. The mark may go up, down, or stay the same.

2. Regrade requests must be made within seven days of the mark publication and sent by email (only) to ac@dal.ca. Include in the subject line *CSCI x193 - Regrade request*.
3. Regrades will be completed before the end of the term in the order they were received.

G. Copyrighted materials

Lectures and distributed instructional materials are protected by copyright. You must have the instructor's permission before recording class lectures or photographing course slides. All materials provided during the term may be used for your study in this course but not otherwise outside the course.

COURSE TOPICS

We follow the three main stages of innovation: value creation, value capture, and value exchange. Topics discussed throughout the term are as follows:

- Thinking Like An Innovator
- Jobs to be Done Theory
- Design Thinking
- Business Models & Customer Models
- Collaborating With Synthetic Customers
- The Value Proposition
- Lean & Agile Methods
- Rapid Prototyping & The MVP
- Getting & Giving Feedback
- IP, Trade Secrets, NDA's
- Sources & Uses of Cash In A Startup
- Pitching to Investors

LEARNING OUTCOMES

- Experience the challenges and rewards of leveraging the different expertise, perspectives and communication styles present on interdisciplinary teams.
- Develop the primary attributes of systematic innovation methods such as design thinking, entrepreneurial mindset, user experience design, and iterative prototyping.
- Apply systematic innovation methods and frameworks to generate and validate human-centred, technology-based solutions to design challenges.
- Gain practical working knowledge of specific communication, version control and project management software tools to aid collaboration and software development.
- Develop a working prototype that solves a challenge in a feasible, viable, and human-centred manner.

ACADEMIC STANDARDS

Failure to correctly attribute sources in your work will result, minimally, in a mark being reduced. For example, not using quotes for text taken from other sources, failing to include in-text citations, or omitting correct information in the references.

Overuse of third-party content, even if adequately attributed, will minimally result in grade reduction for insufficient originality. Additionally, AI-generated text should not be used, even as a cited source or reference, as it does not constitute reliable, verifiable evidence.

If it appears your submission presents other people's work as your own, such as having substantial similarities without correct quotations, citations, and references, your work will be submitted for review by a Faculty Academic Integrity Officer (FAIO). The FAIO will provide an opportunity to talk with them as part of their investigation. FAIO findings have a variety of consequences, including course failure.

These academic standards apply to every item you submit to count toward your course grade.

STUDENT HEALTH AND WELLNESS

Taking care of your health is important. As a Dalhousie student, you have access to a wide range of resources to support your health and wellbeing. Students looking to access physical or mental health & wellness services at Dalhousie can go to the Student Health & Wellness Centre in the LeMarchant Building. The team includes registered nurses, doctors, counsellors and a social worker. Learn more and book an appointment today by [visiting their site](#).

Students also have access to a variety of online mental health resources, including telephone/texting counselling and workshops/training programs. Learn more and [access these resources on this page](#).

CULTURE OF RESPECT^[1]

Every person has a right to respect and safety. We believe inclusiveness is fundamental to education and learning. Misogyny and other disrespectful behaviour in our classrooms, on our campus, on social media, and in our community is unacceptable. As a community, we must stand for equality and hold ourselves to a higher standard.

What we all need to do:

1. **Be Ready to Act:** This starts with promising yourself to speak up to help prevent it from happening again. Whatever it takes, summon your courage to address the issue. Try to approach the issue with open-ended questions like "Why did you say that?" or "How did you develop that belief?"
2. **Identify the Behaviour:** Use reflective listening and avoid labeling, name-calling, or assigning blame to the person. Focus the conversation on the behaviour, not on the person. For example, "The comment you just made sounded racist, is that what you intended?" is a better approach than "You're a racist if you make comments like that."
3. **Appeal to Principles:** This can work well if the person is known to you, like a friend, sibling, or co-worker. For example, "I have always thought of you as a fair-minded person, so it shocks me when I hear you say something like that."

4. **Set Limits:** You cannot control another person's actions, but you can control what happens in your space. Do not be afraid to ask someone "Please do not tell racist jokes in my presence anymore" or state "This classroom is not a place where I allow homophobia to occur." After you have set that expectation, make sure you consistently maintain it.
5. **Find or be an Ally:** Seek out like-minded people that support your views, and help support others in their challenges. Leading by example can be a powerful way to inspire others to do the same.
6. **Be Vigilant:** Change can happen slowly, but do not let this deter you. Stay prepared, keep speaking up, and do not let yourself be silenced.

[\[1\]](https://www.dal.ca/about/mission-vision-values/equity-diversity-inclusion-and-accessibility/about-office-equity-inclusion.html) To find out more, visit The Office of Equity and Inclusion website where you can read about the three separate but integrated units, Human Rights and Equity Services (HRES), Mi'kmaq and Indigenous Relations, and Equity, Diversity, Inclusivity and Accessibility (EDIA): <https://www.dal.ca/about/mission-vision-values/equity-diversity-inclusion-and-accessibility/about-office-equity-inclusion.html>

UNIVERSITY POLICIES

This course is governed by the academic regulations outlined in the **University Calendar**. Dalhousie University has a clear policy on several important issues. Students are strongly advised to familiarize themselves with these policies.

■ Territorial Acknowledgement

Dalhousie University is located in Mi'kma'ki, the ancestral and unceded territory of the Mi'kmaq. We are all Treaty people.

■ Internationalization

At Dalhousie, 'thinking and acting globally' enhances the quality and impact of education, supporting learning that is "interdisciplinary, cross-cultural, global in reach, and orientated toward solving problems that extend across national borders."
<https://www.dal.ca/about-dal/internationalization.html>

■ Academic Integrity

At Dalhousie University, we are guided in our work by the values of academic integrity: honesty, trust, fairness, responsibility, and respect. As a student, you are required to demonstrate these values in all of the work you do. The University provides policies and procedures that every university community member must follow to ensure academic integrity. (read more: http://www.dal.ca/dept/university_secretariat/academic-integrity.html)

■ Accessibility

The Student Accessibility Centre is Dalhousie's centre of expertise for student accessibility and accommodation. If aspects of the design, instruction, and experiences within this course (online or in-person) result in barriers to your inclusion, please contact https://www.dal.ca/campus_life/academic-support/accessibility.html for all courses offered by Dalhousie except Truro.

■ **Conduct in the Classroom — Culture of Respect**

Substantial and constructive dialogue on challenging issues is essential in academic inquiry and exchange. It requires a willingness to listen and tolerance of opposing points of view. All class members must consider individual differences and alternative viewpoints towards each other, instructors, and guest speakers. While expressions of differing perspectives are welcome and encouraged, the words and language used should remain within acceptable bounds of civility and respect.

■ **Diversity and Inclusion — Culture of Respect**

Every person at Dalhousie has a right to be respected and safe. We believe inclusiveness is fundamental to education. We stand for equality. Dalhousie is strengthened by our diversity. We are a respectful and inclusive community. We are committed to being a place where everyone feels welcome and supported, which is why our Strategic Direction prioritizes fostering a culture of diversity and inclusiveness (Strategic Priority 5.2). (read more: <http://www.dal.ca/cultureofrespect.html>)

■ **Student Code of Conduct**

Everyone at Dalhousie is expected to treat others with dignity and respect. The Code of Student Conduct allows Dalhousie to take disciplinary action if students don't follow this community expectation. When appropriate, code violations can be resolved reasonably and informally—perhaps through a restorative justice process. If an informal resolution can't be reached or would be inappropriate, procedures exist for formal dispute resolution. (read more: https://www.dal.ca/dept/university_secretariat/policies/student-life/code-of-student-con.html)

■ **Fair Dealing Policy**

The Dalhousie University Fair Dealing Policy guides the limited use of copyright-protected material without the risk of infringement and without seeking the permission of copyright owners. It is intended to balance the rights of creators and the rights of users at Dalhousie. (read more: https://www.dal.ca/dept/university_secretariat/policies/academic/fair-dealing-policy-.html)

■ **Originality Checking Software**

The course instructor may use Dalhousie's approved originality checking software and Google to check the originality of any work submitted for credit by the Student Submission of Assignments and Use of Originality Checking Software Policy. Without penalty of grade, students are free to choose an alternative method of attesting to the authenticity of their work and must inform the instructor no later than the last day to add/drop classes of their intent to choose an alternate method. (read more: https://www.dal.ca/dept/university_secretariat/policies/academic/student-submission-of-assignments-and-use-of-originality.html)

■ **Student Use of Course Materials**

These course materials are designed for use as part of the CSCI courses at Dalhousie University and are the instructor's property unless otherwise stated. Third-party copyrighted materials (such as books, journal articles, music, videos, etc.) have either been licensed for use in this course or fall under an exception or limitation in Canadian Copyright law. Copying this course material for distribution (e.g., uploading material to a commercial third-party website) may violate copyright law.

■ **Learning and Support Resources**

Find more resources for academic support [*from this page on dal.ca.*](#)